

INFOE 3: LAB 4_ Generalization

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BS INFO 3C

```
1 name,age,zip_code,salary,medical_risk
2 Brian Jones,80,581XX,44162,Low
3 Elizabeth Perez,20,894XX,122257,Low
4 Nicole Gonzalez,40,466XX,127257,Low
5 Gina Walters,60,689XX,9854,High
6 Jessica King,70,716XX,14588,Medium
7 Sonya Chavez,50,561XX,69752,High
8 Matthew Perez,10,195XX,4789,Medium
9 Christine Yates,60,619XX,14845,High
10 Nicole Dickerson,50,164XX,12455,Medium
11 Ian Gibbs,40,430XX,17417,Low
12 John Gibson,40,427XX,1257,Medium
```

- This screenshot displays the successful execution of the generalization script. The output in `generalized_employee_data.csv` shows that quasi-identifiers have been transformed: ages are rounded to the nearest decade and zip codes are masked to the 123XX format, effectively preventing re-identification through linking attacks while preserving the accuracy of sensitive salary data.

The screenshot shows a Visual Studio Code editor window with the following components:

- Explorer:** Displays the file structure of the project, including `employee_data.csv`, `generalized_employee_data.csv`, and `lab4.py`.
- Editor:** Contains the Python script `lab4.py`. The script performs the following steps:
 - Imports `csv` and `copy` modules.
 - Loads the original data from `employee_data.csv` into `raw_data` using `csv.DictReader`.
 - Creates a deep copy of `raw_data` into `new_data` and initializes `total_salary = 0`.
 - Iterates over `new_data` to calculate the total salary and perform generalization:
 - Age Generalization:** Rounds the age down to the nearest 10 (e.g., 82 becomes 80).
 - Zip Code Generalization:** Masks the last two digits of the zip code by keeping the first three and appending 'XX'.
- Terminal:** Shows the command `python lab4.py` being executed. The output includes:
 - A comparison view showing the original and generalized records for 'Brian Jones'.
 - The calculated **Average Salary: 83209.5**.
 - A confirmation message: **Verification: Data successfully generalized.**

- This screenshot demonstrates the implementation of **decade-based generalization** and data integrity verification. The Python script rounds ages down to the nearest decade and masks **zip codes by using string slicing** to keep the first three digits and appending 'XX' to hide the last two. Additionally, the console output displays the **Average Salary**, confirming that the sensitive financial data was preserved accurately throughout the generalization process.

The screenshot shows a Visual Studio Code editor window with the following components:

- EXPLORER:** Displays the file structure for 'LAB4- FRANCISCO, MANULAT, LABANON, CORTEJO...'. It includes files like 'employee_data.csv', 'generalized_employee_data.csv', and 'lab4.py'.
- LAB4.py:** The main script being edited. It contains Python code that:
 - Imports 'csv' and 'copy'.
 - Loads 'employee_data.csv' into 'raw_data' using 'csv.DictReader'.
 - Creates a deep copy of 'raw_data' into 'new_data'.
 - Iterates through 'new_data' to calculate 'total_salary' and categorize 'age' into 'Under 30', '30 to 50', or 'Over 50'.
- TERMINAL:** Shows the execution of 'python lab4.py'. The output includes:
 - 'BEFORE:' data for 'Brian Jones' with age '80'.
 - 'AFTER:' data for 'Brian Jones' with age 'Over 50'.
 - 'Average Salary: 83209.5'.
 - 'Generalized CSV file created successfully.'
- RIGHT SIDEBAR:** Features a 'Build with Agent' section with a warning that 'AI responses may be inaccurate' and a link to 'Generate Agent Instructions'. Below it is a chat interface for 'lab4.py' with a prompt 'Describe what to build'.

- This screenshot demonstrates the final implementation of the 'Life Stages' generalization task. By using conditional logic to categorize ages into 'Under 30,' '30 to 50,' and 'Over 50,' we achieve a higher level of data abstraction. The terminal output confirms the transformation and displays the Average Salary, verifying that the dataset remains useful for analysis while significantly reducing the risk of re-identification.

FileEditSelectionViewGoRunTerminalHelp

Lab4- Francisco, Manulat, Labanon, Cortejo 3C-Infoe3

Update

lab4.pygeneralized_employee_data.csvemployee_data.csv

LAB4- FRANCISCO, MANULAT, LABANON, CORTEJO...

employee_data.csvgeneralized_employee_data.csvlab4.py

lab4.py

15for record in new_data:
16 record["salary"] = record["salary"]
17
18 # Age Category Logic (Extra Task)
19 age = int(record["age"])
20 if age < 30:
21 record["age"] = "Under 30"
22 elif age <= 50:
23 record["age"] = "30 to 50"
24 else:
25 record["age"] = "Over 50"
26
27 # Zip Code Masking
28 record["zip_code"] = record["zip_code"][:3] + "XX"
29
30 # 3. Print Comparison for SS3
31 print("-" * 30)
32 print("COMPARISON VIEW")
33 print("-" * 30)
34 print(f"ORIGINAL RECORD: {raw_data[0]}")
35 print(f"GENERALIZED RECORD: {new_data[0]}")
36 print("-" * 30)
37 print(f"Average Salary: {total_salary / len(new_data)}")
38 print("Verification: Data successfully generalized.")

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

PS D:\Lab4- Francisco, Manulat, Labanon, Cortejo 3C-Infoe3> python lab4.py
ORIGINAL RECORD: {'name': 'Brian Jones', 'age': '80', 'zip_code': '58131', 'salary': '44162', 'medical_risk': 'Low'}
GENERALIZED RECORD: {'name': 'Brian Jones', 'age': 'Over 50', 'zip_code': '581XX', 'salary': '44162', 'medical_risk': 'Low'}

Average Salary: 83209.5
Verification: Data successfully generalized.
PS D:\Lab4- Francisco, Manulat, Labanon, Cortejo 3C-Infoe3>

> OUTLINE
> TIMELINE

Build with Agent
AI responses may be inaccurate
[Generate Agent](#)
[Instructions](#) to onboard AI onto your codebase.

lab4.py
Describe what to build
+ 0 ...
Local 0 Defa...

Ln 38, Col 54 Spaces: 4 UTF-8 CRLF Python Go Live